Pressure & Forces

Exam code: 0625 & 0972

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Pressure

What is pressure?

- **Pressure** is defined as

The force per unit area

- Pressure is measured in pascals (Pa)
  - 1 pascal is equivalent to 1 newton per metre squared
  - 1 Pa = 1 N/m^2
- It can be calculated using the following pressure equation:

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

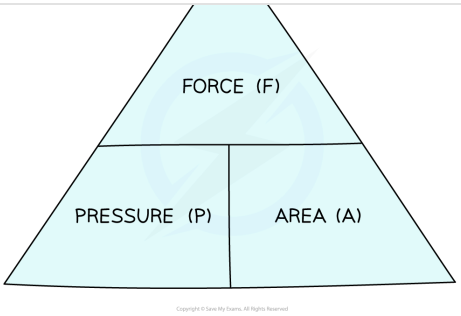
$$p = \frac{F}{A}$$

- Where:
  - p = pressure measured in pascals (Pa) or newtons per metre (N/m^2)
  - F = force measured in newtons (N)
  - A = area measured in metres squared (m^2)
- This equation can be rearranged with the help of a formula triangle:

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



A formula triangle can help rearrange the pressure equation

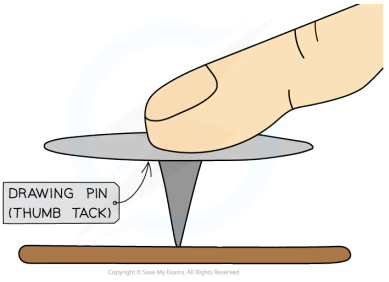
- For more information on how to use a formula triangle refer to the revision note on speed & velocity
- This equation tells us that:
  - If a force is spread over a **large** area it will result in a **small** pressure
  - If it is spread over a **small** area it will result in a **large** pressure

Examples of applications of pressure

- **Example 1:** Tractors
  - Tractors have **large tyres**
  - This spreads the weight (force) of the tractor over a large area
  - This reduces the pressure which prevents the heavy tractor from sinking into the mud

- **Example 2:** Drawing pins
- Drawing pins have **sharp pointed ends** with a very small area
- This concentrates the force, creating a large pressure over a small area
- This allows the drawing pin to be pushed into a wall

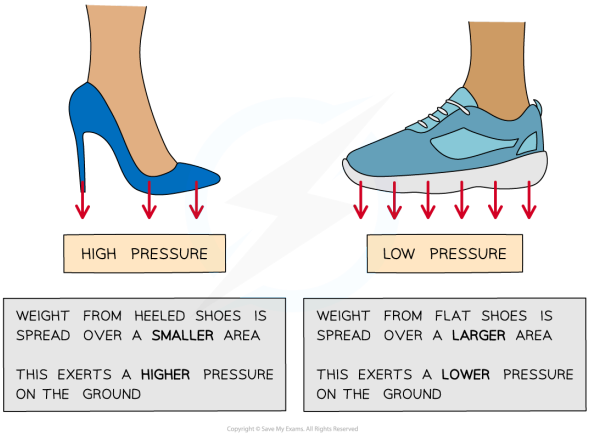
Applying a force to a drawing pin



When you push a drawing pin, it goes into the surface (rather than your finger)

- **Example 3:** High heels
  - High heels have small, sharp points with a small area
  - This concentrates the weight (force), creating a large pressure over a small area
  - Flat shoes have a larger area which the weight (force) is spread over resulting in a lower pressure

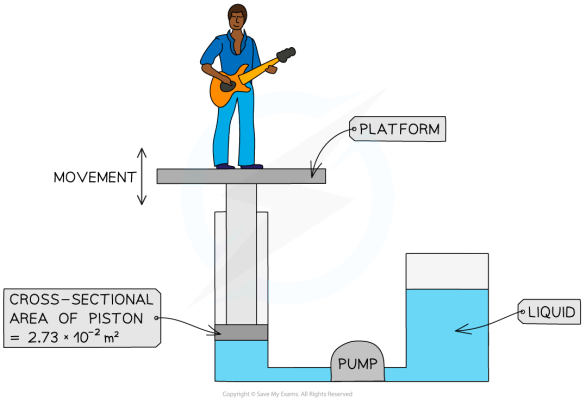
The effect of surface area on pressure



High heels produce a higher pressure on the ground because of their smaller area, compared to flat shoes

Worked Example

The diagram below shows the parts of the lifting machine used to move the platform up and down.



The pump creates pressure in the liquid of 5.28×10^5 Pa to move the platform upwards.

Calculate the force that the liquid applies to the piston.

Answer:

Step 1: List the known quantities

- Cross-sectional area = 2.73×10^{-2} m^2
- Pressure = 5.28×10^5 Pa

Step 2: Write down the relevant equation

Step 3: Rearrange for the force, F

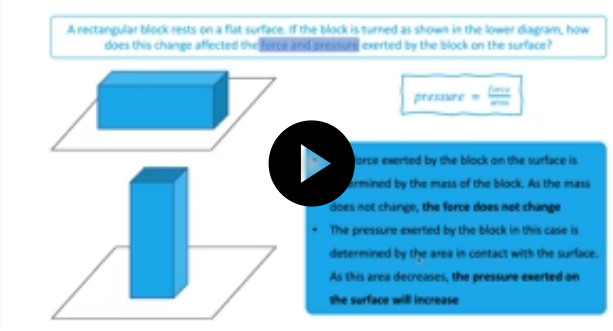
$$F = p \times A$$

Step 4: Substitute the values into the equation

$$F = (5.28 \times 10^5) \times (2.73 \times 10^{-2}) = 14\,414.4$$

Step 5: Round to the appropriate number of significant figures and quote the correct unit

$$F = 14\,400\ N = 14.4\ \text{kN}\ (3\ \text{s.f})$$



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